A Friendly Guide to Loudspeakers: From Small Signal T/S Parameters to Breakup Physics and Beyond

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Overview

This document provides a stepwise approach to loudspeaker modeling with clear explanations on what is assumed, what is added, and how the T/S framework scales toward full-featured loudspeaker modeling. Throughout we adopt the Laplace-domain notation (complex frequency variable s) except where explicitly stated otherwise.

1. **Thiele–Small foundation:** Start with the small-signal T/S model, relating input voltage $V_{\text{in}}(s)$ to on-axis sound pressure $p(s)$ via lumped electrical and mechanical parameters.
2. **Explicit assumptions:** Identify linearity, rigid-piston behavior, infinite baffle conditions, and far-field observation as the baseline simplifications.
3. **Scaling to reality:** Extend the model to include voice-coil inductance, radiation loading, and directivity effects.
4. **Generalization:** Reformulate in a modular, simulation-ready form that can be applied to arbitrary geometries, finite baffles, and code implementations.
5. **Practical Code Implementations:** Implement the derivations into an easy to use form to extract TS++ parameters for a given driver and for a given parameter list produce performance figures.

Notation & Assumptions

- Linear, time-invariant small-signal behavior unless otherwise noted.
- Constant motor strength $B\ell$ (no Bl(x) nonlinearity) for small excursions.
- Infinite, rigid baffle for "baffled piston" analytic formulas (half-space radiation).
- Far-field (Fraunhofer) approximations use $r \gg a^2/\lambda$ unless near-field terms are explicitly kept.
- Laplace domain: s is used everywhere (if you want steady-state set $s \mapsto j\omega$).

1 Symbol Table (select)

Symbol	Meaning	Units
$V_{\text{in}}(s)$	input voltage	V
$I(s)$	coil current	A
$Z_e(s)$	electrical impedance (coil + electromechanical reflection)	Ω
$B\ell$	force factor (force per unit current)	$\mathrm{N\,A^{-1}}$
S_d	piston radiating area	$\mathrm{m^2}$
a	equivalent piston radius ($S_d = \pi a^2$)	m
M_{ms}	moving mass of driver (including cone/coil)	kg
C_{ms}	mechanical compliance	m/N
R_{ms}	mechanical losses (damping)	N s/m
$Z_{\text{rad}}^{\text{ac}}(s)$	acoustic radiation impedance (pressure / velocity)	Pa s/m
$Z_{\text{rad}}^{\text{mech}}(s)$	mechanical radiation impedance (force / velocity)	N s/m
k	acoustic wavenumber (Laplace) $k = \dfrac{s}{c}$	$\mathrm{m^{-1}}$
ρ_0	air density	$\mathrm{kg/m^3}$
c	speed of sound	m/s

2 Impedance Definitions

A. Electrical Impedance

Electrical impedance describes how the voice coil of the driver resists and reacts to an applied voltage across frequency. It is the combination of the coil's inherent resistance and inductance, along with the reflected impedance of the driver's moving system through the force factor (Bl). This reflection links the mechanical domain back into the electrical domain, meaning any change in the cone's motion affects the measured electrical response.

$$Z_e(s) = R_e + sL_e + \frac{(B\ell)^2}{Z_m^{\text{tot}}(s)}$$

where $Z_m^{\text{tot}}(s) = Z_m(s) + Z_{\text{rad,mech}}(s)$ is the total mechanical impedance.

B. Mechanical (Motional) Impedance

Mechanical impedance expresses how the driver's moving assembly resists motion when a force is applied. It combines mechanical resistance Rms (losses in suspension and surround), mass Mms (inertia), and compliance Cms (spring-like suspension). This impedance governs how easily the cone can accelerate or decelerate in response to the input signal, and forms the bridge between the electrical input and acoustic output.

$$Z_m(s) = R_{\text{ms}} + sM_{\text{ms}} + \frac{1}{sC_{\text{ms}}}$$

C. Radiation Impedance (Mechanical and Acoustic Domain) from Rayleigh

This section starts from the Rayleigh (Kirchhoff) surface integral and follows a single logical path to the practical radiation models used in loudspeaker work: general operator, monopole limit, baffled-piston exact form, small-ka series (showing the algebraic step that yields the LF resistance and added mass), far-field algebraic simplification, conversion to mechanical radiation impedance, and finally the numerical ring-BEM fallback. All expressions are given in the Laplace domain with $k = s/c$.

2.1 Rayleigh integral — general acoustic operator.

For an arbitrary radiating surface S with surface-normal velocity $v(\mathbf{r}', s)$ the Laplace-domain Rayleigh integral gives the pressure at a field point $\mathbf{r}$:

$$\boxed{p(\mathbf{r}, s) = \frac{s\rho_0}{2\pi} \iint_S v(\mathbf{r}', s) \frac{e^{-k|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} \, dS'} \qquad \left(k = \tfrac{s}{c}\right).$$

Define the average surface velocity and radiator area

$$v_{\text{avg}}(s) = \frac{1}{S_d} \iint_S v(\mathbf{r}', s) \, dS', \qquad S_d = \iint_S dS'.$$

Then the acoustic velocity-to-pressure operator at $\mathbf{r}$ (an impedance-like operator) is

$$\boxed{Z_{\text{rad}}^{\text{ac}}(s; \mathbf{r}) = \frac{p(\mathbf{r}, s)}{v_{\text{avg}}(s)} = \frac{s\rho_0}{2\pi} \frac{1}{v_{\text{avg}}(s)} \iint_S v(\mathbf{r}', s) \frac{e^{-k|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} \, dS'.}$$

2.2 Mechanical radiation impedance (force/velocity).

The mechanical radiation impedance (which enters the motional branch of the driver) is the total radiated force per (average) velocity:

$$\boxed{Z_{\text{rad}}^{\text{mech}}(s) = \frac{F_{\text{rad}}(s)}{v_{\text{avg}}(s)} = \frac{1}{v_{\text{avg}}(s)} \iint_S p_{\text{near}}(\mathbf{r}', s) \, dS'}$$

(units N·s/m). Under the uniform-face-pressure approximation (rigid piston with nearly uniform pressure) the two operators are related simply by area:

$$\boxed{Z_{\text{rad}}^{\text{mech}}(s) \approx S_d\, Z_{\text{rad}}^{\text{ac}}(s)}$$

(i.e. force ≈ pressure × area). Be explicit which form you use when assembling $Z_m^{\text{tot}}(s)$.

2.3 Monopole (spherical) limit — small radiator, far-field.

If the radiator is compact relative to the wavelength the total volume velocity is

$$Q(s) = \iint_S v(\mathbf{r}', s)\, dS' = S_d\, v_{\text{avg}}(s),$$

and the Laplace-domain far-field on-axis pressure of a point monopole at distance r is

$$\boxed{p_{\text{mono}}(r, s) = \rho_0\, s\, \frac{Q(s)}{4\pi r}\, e^{-sr/c} = \rho_0\, s\, \frac{S_d}{4\pi r}\, v_{\text{avg}}(s)\, e^{-sr/c}}$$

so the monopole acoustic impedance (velocity-to-pressure) is

$$\boxed{Z_{\text{mono}}^{\text{ac}}(s, r) = \rho_0\, s\, \frac{S_d}{4\pi r}\, e^{-sr/c}.}$$

This is causal (contains $e^{-sr/c}$) and useful as the small-source algebraic approximation.

2.4 Baffled circular piston — exact on-axis (Bessel/Hankel) form.

For a rigid circular piston of radius a in an infinite rigid baffle the Rayleigh integral evaluates on-axis to a closed-form involving Bessel functions. With $S_d = \pi a^2$ and $k = s/c$:

$$\boxed{Z_{\text{piston}}^{\text{ac}}(s) = \rho_0 c S_d \Big(1 - \frac{J_1(2ka)}{ka}\Big) \;+\; \rho_0 c S_d \Big(\frac{2}{\pi ka} Y_1(2ka)\Big)}$$

This is the exact (on-axis) acoustic velocity-to-pressure operator as a complex function of s. On the $s = j\omega$ axis the first bracket produces the resistive contribution and the second bracket the reactive contribution.

Numerical note: evaluate the exact Bessel/Hankel expression carefully for small $|ka|$. Although not singular, the representation contains terms with $1/(ka)$ that cause cancellation and loss of precision as $ka \to 0$. Use series expansions (Taylor) for $|ka| \ll 1$ or robust special-function libraries that support complex arguments.

2.5 Small-ka (Taylor) expansion — algebraic derivation of LF resistance and added mass.

We now show how the exact piston form reduces to the familiar low-frequency lumped expressions by algebraic expansion.

Start with the small-argument expansion for $J_1(x)$ (with $x = 2ka$):

$$J_1(x) = \frac{x}{2} - \frac{x^3}{16} + O(x^5).$$

Hence

$$\frac{J_1(2ka)}{ka} = 1 - \frac{(ka)^2}{2} + O((ka)^4).$$

Substitute this into the first bracket of the exact piston expression:

$$\rho_0 c S_d \Big(1 - \frac{J_1(2ka)}{ka}\Big) \approx \rho_0 c S_d \cdot \frac{(ka)^2}{2}.$$

Thus the leading low-frequency radiation resistance is

$$\boxed{R_{\mathrm{rad}}(s) \approx \rho_0 c S_d \, \frac{(ka)^2}{2} = \rho_0 c S_d \, \frac{(sa/c)^2}{2} \, .}$$

The leading reactive (near-field / added-mass) term comes from the matched expansion of the second (Hankel/Y_1) term. Carrying out this matched-series expansion (standard result) yields

$$\boxed{X_{\mathrm{rad}}(s) \approx \rho_0 c S_d \, \frac{8\,ka}{3\pi} = \rho_0 S_d \frac{8a}{3\pi} \, s \, .}$$

Therefore the low-frequency added mass (kg) is

$$\boxed{M_{a,\mathrm{LF}} = \frac{X_{\mathrm{rad}}(s)}{s} = \rho_0 S_d \frac{8a}{3\pi} = \frac{8}{3} \rho_0 a^3 \, .}$$

(Equivalently $M_a = \rho_0 S_d \varepsilon_{\mathrm{LF}}$ with $\varepsilon_{\mathrm{LF}} = 8a/(3\pi)$.)

2.6 Where the LF expansion breaks down (practical guidance).

- The $(ka)^2/2$ resistance and $8ka/(3\pi)$ reactance are the first nonzero terms of the Bessel/Hankel expansion and are valid only for $|ka| \ll 1$.
- Conservative rule-of-thumb: $|ka| \lesssim 0.3$ for errors $\lesssim 10\%$. For $ka \sim O(1)$ diffractive and directivity effects dominate; higher-order oscillatory terms from the exact expression become important.
- The reactive (added-mass) term is near-field stored energy and shifts the mechanical resonance f_s; incorrect added-mass factors change low-frequency SPL / resonance predictions.

2.7 Far-field on-axis algebraic simplification (baffled case).

When the observer is in the far field (Fraunhofer condition; typically $r \gg a^2/\lambda$), the spatial integral approximately factors and the on-axis pressure becomes monopole-like with the half-space prefactor (baffle):

$$p_{\mathrm{piston,on\text{-}axis}}(r, s) \approx \rho_0 \, s \, \frac{S_d}{2\pi r} \, v_{\mathrm{avg}}(s) \, e^{-sr/c}.$$

Hence the far-field baffled-on-axis acoustic impedance is

$$\boxed{Z^{\mathrm{ac}}_{\mathrm{baffled,on\text{-}axis}}(s, r) = \rho_0 \, s \, \frac{S_d}{2\pi r} \, e^{-sr/c} \, .}$$

(Physically: full-space monopole has denominator $4\pi r$; the baffle restricts radiation to half-space giving the $2\pi r$ factor.)

When it is safe to drop $e^{-sr/c}$ like in the classic TS model? Dropping the exponential is acceptable when:

- the model targets low frequencies where the phase delay across the band is negligible, or
- the application is sensitivity-level estimation, low-order control design, or steady-state SPL prediction where exact phase is not required, and
- r is fixed and large enough that the delay manifests mainly as an (almost) constant group delay over the modeled band.

If causal time-domain accuracy or controller-phase matching is required, keep $e^{-sr/c}$ or substitute a Padé approximation.

2.8 Converting acoustic operator to mechanical radiation impedance for motional branch.

If the mechanical model uses the integrated radiated force, under uniform-face-pressure

$$Z_{\text{rad}}^{\text{mech}}(s) \approx S_d \, Z_{\text{rad}}^{\text{ac}}(s).$$

Therefore inserting the far-field algebraic form into the mechanical branch gives (delay dropped if desired)

$$Z_{\text{rad,mech}}(s) \approx S_d \cdot \frac{\rho_0 s S_d}{2\pi r} = \frac{\rho_0 s S_d^2}{2\pi r},$$

which shows explicitly the S_d^2 scaling in the mechanical domain when using an acoustic numerator proportional to S_d.

2.9 Ring-based numerical BEM (full ka, arbitrary $v(r)$).

For accuracy at all ka, for non-uniform velocity, or for finite-baffle/array interactions discretize the radiator into N rings (areas dS_i) and use the causal Laplace-domain Green's function $G(r_i, r_j, \theta) = e^{-kR_{ij}}/R_{ij}$ with

$$R_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta}.$$

The Laplace-domain mutual mechanical impedance between rings i, j (with M-point angular quadrature) is

$$Z_{ij}(s) \approx \frac{s\rho_0}{4\pi} \, dS_i \, dS_j \, \frac{1}{M} \sum_{m=1}^{M} \frac{e^{-kR_{ij,m}}}{R_{ij,m}},$$

and

$$Z_{\text{rad,BEM}}(s) = \sum_{i=1}^{N} \sum_{j=1}^{N} Z_{ij}(s), \qquad M_a(s) = \frac{\Im\{Z_{\text{rad,BEM}}(s)\}}{s}.$$

2.10 Implementation and numerical notes

- Treat the weak self-term ($i = j$) analytically or with refined quadrature; verify convergence by mesh refinement.
- Evaluate Bessel/Y functions for complex arguments with a robust special-function library when using the exact piston form.
- When building rational approximants (Padé, vector fitting) for time-domain simulation compute on the Laplace axis so causality is preserved.
- Explicitly state where you drop $e^{-sr/c}$ (delay) — mixing delay-free numerators with high-fidelity mechanical $Z_m^{\text{tot}}(s)$ is permissible but must be justified.

2.11 Key Takeaways

- **Constant Model:** Best for low frequencies; very simple and fast.
- **Bessel Function Model:** Suitable for high frequencies; moderate complexity.
- **BEM Model:** Accurate across all frequencies; moderate complexity.

3 Unified Transfer Function Derivation

3.1 Voltage-to-pressure (closing the loop)

Using the electromechanical coupling the (average) cone velocity is

$$v_{\text{avg}}(s) = \frac{B\ell}{Z_m^{\text{tot}}(s)}\, I(s) = \frac{B\ell}{Z_m^{\text{tot}}(s)} \cdot \frac{V_{\text{in}}(s)}{Z_e(s)},$$

so that the voltage-to-velocity transfer is

$$H_v(s) \triangleq \frac{v_{\text{avg}}(s)}{V_{\text{in}}(s)} = \frac{B\ell}{Z_m^{\text{tot}}(s)\, Z_e(s)}.$$

The acoustic pressure at the listening point (for any of the three levels above) is obtained by applying the appropriate acoustic impedance:

$$p(s) = Z_{\text{rad}}^{\text{ac}}(s)\; v_{\text{avg}}(s),$$

hence the voltage-to-pressure transfer is

$$H_p(s) \triangleq \frac{p(s)}{V_{\text{in}}(s)} = Z_{\text{rad}}^{\text{ac}}(s)\; H_v(s).$$

Substituting the baffled piston far-field algebraic form and (optionally) dropping the exponential delay term gives the classical TS model. Note we consistently use s (Laplace variable) throughout:

$$\boxed{H_p(s) \;=\; \frac{s \cdot \dfrac{B\ell\,\rho_0 S_d}{2\pi r}}{(B\ell)^2 + \big(R_e + sL_e\big)\; Z_m^{\text{tot}}(s)}}$$

where

$$\boxed{Z_m^{\text{tot}}(s) = s\big(M_{\text{ms}} + M_a(s)\big) + R_{\text{ms}} + \frac{1}{sC_{\text{ms}}} + Z_{\text{rad}}^{\text{mech}}(s)}$$

and $Z_{\text{rad}}^{\text{mech}}(s)$ should be formed consistently (e.g. via $Z_{\text{rad}}^{\text{mech}}(s) \approx S_d Z_{\text{rad}}^{\text{ac}}(s)$ under the uniform-pressure approximation, or from a BEM-derived surface integral for higher fidelity). Use $M_a(s)$ to indicate the (in general) frequency-dependent added mass unless intentionally employing a low-frequency constant M_a approximation.

Practical modeling notes.

- Use the full Rayleigh integral when per-element accuracy or non-uniform face velocity is important (evaluate numerically via ring-BEM or panel BEM).
- Use the monopole form when the radiator is electrically/physically small relative to wavelengths of interest.
- Use the baffled far-field algebraic factor for simple on-axis sensitivity and control-oriented models; include Padé approximations for the causal delay if you need a rational model.
- When converting between acoustic and mechanical impedances be careful: $Z_{\text{rad}}^{\text{mech}}(s)$ (N·s/m) relates to $Z_{\text{rad}}^{\text{ac}}(s)$ (Pa·s/m) through surface-area integration and is only simply $S_d Z_{\text{rad}}^{\text{ac}}(s)$ under the uniform-pressure approximation.

Key physical relationships:

$$\text{Acceleration to Pressure:} \quad p(s) = \underbrace{\frac{\rho_0 S_d}{2\pi r}}_{K_{\text{rad}}}\, s^2 x(s),$$

$$\text{Velocity to Pressure:} \quad p(s) = \underbrace{\frac{\rho_0 S_d s}{2\pi r}}_{Z_{\text{rad}}^{\text{ac}}(s)}\, v(s),$$

$$\text{Position to Pressure:} \quad p(s) = s^2 K_{\text{rad}} x(s).$$

4 Region Overview

4.1 Frequency-Domain Physics and Design Implications

Frequency Range	Dominant Physics	Pressure relation (dominant)	Design Focus
$f < f_s$	Suspension compliance dominates	$p \propto s^2 x$ (radiation small)	- Excursion limits - $X_{\max}$ optimisation - High-compliance suspension
$f_s < f < f_e$	Mass control (inertia)	$p \propto sv$	- Sensitivity optimisation - Motor strength ($B\ell$) - Low M_{ms}
$f > f_e$	Voice coil inductance	electrical rolloff affects drive	- Shorting rings - Copper caps - Crossover compensation
$f > f_{\text{rad}}$	Radiation directivity / lobing	p shows strong ka dependence	- Piston size optimisation - Dome vs cone tweeters - Phase plugs

where

$$f_s = \frac{1}{2\pi\sqrt{(M_{\text{ms}} + M_a)C_{\text{ms}}}} \qquad \text{(Resonance frequency)}$$

$$f_e = \frac{R_e}{2\pi L_e} \qquad \text{(Inductive rolloff, rough)}$$

$$f_{\text{rad}} = \frac{c}{2\pi a} \qquad \text{(frequency at which } ka \sim 1\text{)}$$

$$a = \sqrt{S_d/\pi} \qquad \text{(Equivalent piston radius)}$$

4.2 Speakers in Series (isobaric) and Parallel

Assumption: $r \ll \lambda$ and isobaric cavity mass negligible.

Configuration	**S** (Radiating area)	**C** (Compliance)	$\mathbf{V}_{\text{as}}$ (Eq. volume)	$\mathbf{P}_{\max}$ (Power handling)	ΔL_p (vs. single)
Single driver	S	C	V_{as}	P	0 dB
Isobaric/Series pair	S	$\frac{C}{2}$	$\frac{V_{\text{as}}}{2}$	$2P$	$\begin{cases} 0\,\text{dB}, & P \text{ (same total)} \\ +3\,\text{dB}, & 2P \text{ (doubled)} \end{cases}$
Two drivers in parallel	$2S$	$2C$	$2V_{\text{as}}$	$2P$	$\begin{cases} +3\,\text{dB}, & P \text{ (same total)} \\ +6\,\text{dB}, & 2P \text{ (doubled)} \end{cases}$

Table: Comparison of a single driver, an isobaric pair, and two drivers in parallel (coherent; otherwise -3 dB when incoherent)

4.3 Source Addition in Spherical Speaker Arrays

4.3.1 Principles at the Listening Point

- **Coherent sources:** sources with fixed phase and frequency relationships; sound pressures add linearly: $p_{\text{total}} = N \cdot p_{\text{single}}$. Level increases by $20 \log_{10} N$.
- **Incoherent sources:** intensities add, pressures combine statistically as root-sum-square: $p_{\text{total}} = \sqrt{N}\, p_{\text{single}}$. Level increases by $10 \log_{10} N$.

4.3.2 Arbitrary Speaker Scaling by Solid Angle

For a sound source with known reference SPL $L_{p,\text{ref}}$ at $P_{\text{ref}} = 1\,\text{W}$, $r_{\text{ref}} = 1\,\text{m}$ and reference solid angle $\Omega_0 = 4\pi$ the general scaling is

$$L_p(r, P, \Omega) = L_{p,\text{ref}} + 10 \log_{10} \left(\frac{P}{P_0} \right) - 20 \log_{10} \left(\frac{r}{r_0} \right) + 10 \log_{10} \left(\frac{\Omega_0}{\Omega} \right).$$

Geometry	Solid Angle Ω	Level Gain [dB]
Spherical wave (free field)	4π	0 dB
Half-space (baffle)	2π	+3.01 dB
Quarter-space (corner)	π	+6.02 dB
Eighth-space (floor & corner)	$\frac{\pi}{2}$	+9.03 dB

5 Implementation Overview (github.com/theIvanR/speakeasy)

Two primary objectives motivate the implementation:

1. **Forwards Problem:** Given a set of Thiele–Small parameters (optionally extended with frequency-dependent terms), predict the loudspeaker's electrical, mechanical, and acoustic performance—SPL, displacement, efficiency, impedance, etc.
2. **Reverse Problem:** For an unknown driver, use measured data (e.g., impedance or SPL curves) to identify the T/S parameters, possibly including extended ones such as frequency-dependent added mass or voice-coil inductance.

5.1 Forward Simulation Implementation

The core simulation is split into two files:

- `simulate_speaker_full_example.m` – a demo script that defines frequency grid, driver parameters, and configuration options, then calls the main solver.
- `simulate_speaker_full.m` – the main function that performs the electroacoustic simulation.

5.1.1 Main Function Overview

`simulate_speaker_full` accepts a frequency vector `f`, a parameter structure `p` (containing Thiele–Small data, air properties, and optional box parameters), and an options structure `opts`. It returns a structured result with all computed quantities and optionally generates plots.

Key features:

- **Configurations:** `opts.cfg` selects `'single'`, `'series'` (ideal isobaric), `'parallel'`, or `'series_parallel'`.
- **Enclosures:** infinite baffle (IB) and sealed box are always computed. The sealed box uses a compliance C_{box} derived from volume V_b and an optional damping coefficient α (for lossy box models).
- **Radiation models:** `opts.radiationModel` can be `'constant'` (low-frequency series expansion), `'bessel'` (exact baffled piston), or `'bem'` (numerical ring-BEM via a separate function `pistonBEM`).
- **Electrical and mechanical impedances:** built from the chosen radiation model, including the frequency-dependent added mass $M_a(\omega)$.
- **Derived outputs:** SPL (reference distance r_{ref}), cone displacement, acoustic/electrical power, efficiency, and impedance magnitude/phase.
- **Resonance extraction:** The resonance frequency is found by locating the zero crossing of $\Im\{Z_m^{\text{tot}}\}$ (linear interpolation between grid points). From this, M_a at resonance, effective mass, and Q are computed.
- **Self-consistent box tuning:** For a sealed box, the resonance is iteratively updated because M_a depends on frequency; the routine `estimate_self_consistent_box_resonance` handles this.
- **Plots:** `opts.plot` controls the output – `'none'`, `'reduced'` (combined panel of displacement, SPL, efficiency), or `'full'` (separate figures for SPL, displacement, efficiency, impedance, powers, plus the reduced panel).

5.1.2 Extending to Nonlinear / Time-Varying Behavior

The code is modular; the nonlinear effects mentioned in the text (thermal drift, amplitude-dependent compliance/damping, motor force variation) can be incorporated by replacing the constant parameters in `p` with functions that depend on instantaneous amplitude or temperature. Such extensions are straightforward because the core solver already separates the linear mechanical, electrical, and radiation blocks.

5.2 Inverse Problem Implementation

Extracting Thiele–Small parameters from an unknown driver is traditionally a multi-step ordeal. Commercial solutions are often locked behind expensive paywalls or provide incomplete access to the underlying physics. Academic toolboxes exist, but many are either not publicly available, poorly documented, or rely on outdated measurement techniques. DIY approaches typically involve a mix of:

- **Added-mass methods:** attaching Blu-Tack or modelling clay to the cone to measure the shift in resonance frequency—a procedure that is messy, prone to error, and requires careful mass calibration.
- **Vacuum chambers:** used to measure the driver in a near-vacuum to isolate mechanical parameters, which is impractical for most hobbyists.

- **Separate impedance and SPL measurements:** often requiring anechoic conditions or complex gating, and then manually piecing together parameters from different tests.

These methods are not only cumbersome but also introduce significant uncertainty, especially for extended parameters like voice-coil inductance or frequency-dependent added mass.

Our Approach: One-Shot, Hardware-Simple, Gradient-Based Extraction We take a different route. The forward model described in the previous section is used as the basis for a nonlinear least-squares fit. The only hardware required is:

- A soundcard with at least two channels (e.g., a simple USB audio interface).
- A precision resistor (typically $R_{\text{ref}} \approx 10\,\Omega$ to $100\,\Omega$) connected in series with the driver.
- A computer running MATLAB (or Python) to generate a pertubation and record the response across the driver and the reference resistor.

From these two voltage measurements, the electrical impedance $Z_e(\omega)$ is computed directly in the frequency domain. No added mass, no vacuum, no anechoic chamber—just a simple electrical measurement. This impedance curve contains all the information needed to recover the T/S parameters.

5.2.1 Impedance Measurement with a Soundcard

The measurement is performed by a Python class `BodePlotter` that uses the `sounddevice` library. The key steps are:

1. **Stimulus generation:** A band-limited white noise signal is generated, filtered to the audio band (e.g., 20–20kHz), and scaled to a desired RMS voltage. *Why band-limited white noise?* Unlike a swept sine, noise excites all frequencies simultaneously. This allows the transfer function to be estimated via cross-correlation (Welch's method) over the entire band in a single acquisition. Multiple averages are built into the Welch estimator, which dramatically improves signal-to-noise ratio and rejects uncorrelated disturbances (including nonlinear harmonics and background noise). Moreover, the coherence function becomes available as a direct quality metric—something a swept sine cannot provide without multiple measurements. The measurement is fast, robust against time-variations, and easily repeatable.
2. **Playback and recording:** The stimulus is sent to the output channel; simultaneously two input channels record:
   - Channel 1: voltage across the sense resistor (source side).
   - Channel 2: voltage across the DUT (driver side).
3. **Transfer function estimation:** Using Welch's method, the cross-spectral density P_{yx} and the power spectral density P_{xx} are computed. The transfer function $H = V_{\text{DUT}}/V_{\text{src}}$ is obtained as $H = \overline{P_{yx}}/P_{xx}$ (the conjugate ensures correct phase). This is a frequency-domain representation of the input-output cross-correlation, which effectively averages over many time segments, suppressing noise and nonlinearities that are not correlated with the stimulus.
4. **Impedance calculation:** From the voltage divider equation,

$$Z_{\text{DUT}}(\omega) = R_{\text{sense}} \frac{H(\omega)}{1 - H(\omega)},$$

   where R_{sense} is the known series resistor.
5. **Coherence:** The coherence function $\gamma^2(\omega) = |P_{yx}|^2/(P_{xx}P_{yy})$ is also computed. It indicates the degree of linear correlation between input and output. Frequency bins with low coherence (e.g., < 0.8) are discarded, as they correspond to regions where the measurement is corrupted by noise or nonlinearities. This built-in quality assessment is a key advantage over swept-sine methods.

The resulting impedance magnitude and phase are saved to a CSV file. The same setup can easily be extended for nonlinear measurements (e.g., using Volterra series or Wiener orthonormal bases) by injecting multi-tone or stepped-amplitude signals, and to capture parameter variations with bias current or signal level.

5.2.2 Parameter Fitting in MATLAB

With the measured impedance data, a MATLAB routine fits the T/S parameters using the same forward model `simulate_speaker_full`. The fitting process is:

1. **Data cleaning:** The CSV is read, rows with coherence below a threshold (e.g., 0.8) are discarded, and the data are optionally down-sampled.

2. **Initial guess extraction:** The resonance frequency f_s, peak impedance Z_{peak}, and high-frequency R_e and L_e are estimated directly from the impedance curve. The quality factors Q_{ms} and Q_{es} are derived, providing starting values for Bl, R_{ms}, and M_{ms}.
3. **Nonlinear optimization:** A Levenberg–Marquardt solver (`lsqnonlin`) minimizes the weighted complex residual between measured impedance Z_{meas} and the model impedance $Z_{\text{model}}(\theta)$, where θ comprises the T/S parameters. The residual is defined as

$$r = \left[\frac{\Re\{Z_{\text{model}} - Z_{\text{meas}}\}}{\text{scale}} \cdot w, \ \frac{\Im\{Z_{\text{model}} - Z_{\text{meas}}\}}{\text{scale}} \cdot w \right]^{\mathsf{T}},$$

   with scale $= \max(|Z_{\text{meas}}|, 0.25\, R_e)$ and w a weight proportional to coherence.
4. **Parameter extraction:** The optimization returns a complete set of T/S parameters, including S_d (if not already known) and the extended parameters L_e and R_{ms}. The result is a "TS++" parameter set that can be used directly for enclosure design, crossover simulation, and nonlinear extensions.

The entire extraction is performed in one shot, using only a simple impedance measurement. The code is open and modular, making it easy to adapt for different driver configurations or to incorporate additional physics (e.g., frequency-dependent inductance, mechanical nonlinearities). Because the same forward model is used for both simulation and parameter fitting, the extracted parameters are fully consistent with the simulated performance.

Modularity and Beyond: Multi-Driver Systems and Enclosure Measurements The framework is designed from the ground up to be modular. The forward simulation routine `simulate_speaker_full` already accepts configuration options for multiple driver arrangements (`'single'`, `'series'`, `'parallel'`, `'series_parallel'`) and for sealed enclosures (with volume V_b and damping α). This means the same fitting code can be used to characterize a complete loudspeaker system—for example, a pair of drivers in a sealed box—by simply changing the configuration and including the enclosure parameters in the model.

When the system under test contains multiple drivers or a known enclosure, the optimization can directly estimate the effective parameters of the combined system. From these, the individual driver parameters can be recovered if the arrangement and box properties are known. Conversely, if the driver parameters are known, the fitting can be adapted to determine the enclosure volume or the added stiffness of a sealed box by treating V_b (or C_{box}) as an additional free parameter. Minor modifications to the code (adding the box compliance to the mechanical impedance and including V_b in the optimization vector) enable this capability.

The modularity also extends to the measurement side: the same hardware and Python acquisition script can be used to collect impedance data for any passive electroacoustic network. By changing the stimulus parameters (e.g., amplitude, duration) and the post-processing settings, the system can be used for nonlinear characterisation (Volterra series, Wiener orthonormal bases) or for measuring parameter drift under thermal stress—all without rewriting the core simulation engine.

Short reference list

Lord Rayleigh, *The Theory of Sound*, Vol. I (Macmillan, 1878).
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